

CFD COMPASS

An open reference
for fluid dynamics.

150 canonical repositories
14 topic areas
18 official external resources

Reference edition 1.0.0
Snapshot: 2026-10-25

Curated by Khalil | github.com/Khalil-Work/cfd-compass

Linked projects remain the work of their original contributors.

Navigate the reference

Choose a topic below, use the PDF bookmarks, or jump from the alphabetical index to an individual project. Every repository name and documentation label is clickable.

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Read this first

A problem-first directory

This collection helps you discover tools and compare their intended roles. Start with the physical regime, coupled physics, required outputs, geometry and evidence you need. Then consider discretization, compute budget, workflow and adoption constraints.

What checking means

Entry scope was reviewed against primary upstream repositories or documentation. GitHub ownership, licence identifiers, archive flags and activity dates were fetched as a dated metadata snapshot. Listed solvers were not all installed, executed or independently validated.

Understand the labels

Established, research and emerging are editorial context labels. Stars are dated discovery metadata, not accuracy rankings. A repository push is not necessarily a release. NOASSERTION means the API could not identify a standard licence; inspect the actual terms. Read caveats for official mirrors, maintenance notices and source-available restrictions.

Use upstream credit

The software is the work of its original contributors. Follow their licences and cite their publications when using them. This directory provides original catalogue descriptions and discovery tools; it does not redistribute the listed codebases or claim their authorship.

Keep the snapshot in context

This edition records metadata on 2026-10-25. Links, licences and maintenance can change. The live directory and reviewed corrections are available at github.com/Khalil-Work/cfd-compass. Catalogue text is CC BY 4.0; tooling is MIT; upstream terms remain separate.

Choose a stack

Learning and first studies

Start with verification/learning, then general solvers. Check dimensions, boundary conditions, stability and grid convergence.

Aerodynamics and design

Compare general solvers, high-order methods and optimization tools. Confirm flow regime, turbulence modelling and adjoint consistency.

Combustion, sprays and interfaces

Begin with reacting-flow or multiphase tools. Check chemical mechanism validity, stiffness, transport, phase-change assumptions and mass/energy coupling.

Turbulence and scale

Inspect DNS/LES resolution, wall treatment, domain and statistical sampling. Compare memory requirements, parallel support and actual precision/hardware constraints.

Coupled or environmental problems

Confirm the coupling interfaces, conservation, stability, forcing data, constitutive assumptions and domain/grid nesting.

Learned models and inverse studies

Check differentiability assumptions, discrete gradients, training/test separation, out-of-distribution behavior and numerical baselines.

Save a reproducibility record

Record the upstream version, mesh, physical parameters and units, initial/boundary conditions, model choices, tolerances, time step or CFL, hardware/precision, environment, post-processing script, comparison data and uncertainty. Reproduce a quantity of interest, not only a visualization.

General-purpose solvers & frameworks

Finite-volume and finite-element starting points for engineering flow simulation.

11 repositories in this topic.

ADflow

Compressible Euler, laminar Navier-Stokes and RANS solver on structured multiblock and overset meshes, with a discrete adjoint and derivative-verification facilities for aerodynamic design optimization.

For: Gradient-based aerodynamic shape optimization

Established | Fortran | NOASSERTION | Last push: 2026-10-19

mdolab/adflow / Documentation

Selection note: Targets structured and overset design workflows; a mesh generator and the surrounding MDO dependencies may be required for complete optimization studies.

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code_saturne

EDF's general-purpose CFD system for industrial flow and transport studies, providing finite-volume numerics, turbulence and heat-transfer models, parallel execution, and associated case setup tooling.

For: Industrial thermal-fluid simulations

Established | C++ | GPL-2.0 | Last push: 2026-10-22

code-saturne/code_saturne / Documentation

Selection note: The project describes this GitHub repository as its public mirror; follow official release and contribution instructions rather than treating every branch as a release.

COOLFluiD

Object-oriented high-performance platform for computational fluid dynamics, plasma and multiphysics simulations, exposing a modular research environment for developing and combining numerical methods and physical models.

For: CFD and plasma solver development

Research | Prolog | LGPL-3.0 | Last push: 2026-10-23

andrealani/COOLFluiD / Documentation

Selection note: Check the relevant plugin documentation and licensing details; this is a developer-oriented platform with a substantial build configuration.

Elmer FEM

Multiphysics finite-element package combining fluid mechanics, heat transfer and other field solvers, with mesh and graphical tools plus parallel infrastructure for coupled scientific simulations.

For: Coupled finite-element thermal-fluid problems

Established | Fortran | NOASSERTION | Last push: 2026-10-23

ElmerCSC/elmerfem / Documentation

Selection note: Select the solver module, discretization and coupling strategy explicitly; breadth of available physics does not imply every coupling is turnkey.

FiPy

Python finite-volume PDE framework from NIST for assembling coupled convection, diffusion, transient and source terms, particularly useful for transport, phase-field and materials-model prototyping and research.

For: Transport-equation and phase-field prototypes

Established | Python | NOASSERTION | Last push: 2026-10-14

usnistgov/fipy / Documentation

Selection note: A PDE-building framework rather than a complete industrial Navier-Stokes workflow; users must formulate and verify their coupled equations.

Fluidity

Unstructured finite-element and control-volume CFD code with anisotropic mesh adaptivity, parallel execution and Python interfaces, used for multiphase, geophysical and general fluid-dynamics simulations.

For: Anisotropic adaptive flow simulations

Established | Fortran | NOASSERTION | Last push: 2026-10-19

FluidityProject/fluidity / Documentation

Selection note: The dependency and platform setup is substantial; use installation instructions matching the chosen branch and environment.

Kratos Multiphysics

Modular C++ multiphysics framework with Python interfaces and dedicated fluid-dynamics, fluid-structure interaction and particle applications, supporting coupled engineering workflows and parallel simulation development.

For: Coupled fluid-structure engineering applications

Established | C++ | NOASSERTION | Last push: 2026-10-24

KratosMultiphysics/Kratos / Documentation

Selection note: Capabilities belong to specific applications; the core framework alone is not a configured CFD solver.

Lethe

Finite-element CFD and discrete-element project using deal.II, with high-order continuous Galerkin formulations, adaptive meshes and coupled fluid-particle solvers for single-phase and multiphase engineering research.

For: Adaptive CFD and CFD-DEM research

Research | C++ | Apache-2.0 | Last push: 2026-10-24

chaos-polymtl/lethe / Documentation

Selection note: Requires a compatible deal.II dependency stack; choose the relevant solver and reproduce its documented benchmark before extending a model.

MOOSE Navier-Stokes

Navier-Stokes module within the MOOSE multiphysics framework, offering finite-element and finite-volume formulations for pressure-based, compressible and porous-flow applications with shared nonlinear-solver infrastructure and coupling mechanisms.

For: Custom coupled flow applications

Established | C++ | LGPL-2.1 | Last push: 2026-10-24

idaholab/moose / Documentation

Selection note: Read the module's solver-capability matrix: turbulence, two-phase and compressibility support varies between formulations and is actively evolving.

OpenFOAM Foundation

Finite-volume CFD platform with an extensive solver, physical-model and meshing ecosystem, covering incompressible and compressible flows, turbulence, heat transfer, reacting systems and multiphase problems.

For: Extensible engineering CFD workflows

Established | C++ | NOASSERTION | Last push: 2026-10-24

OpenFOAM/OpenFOAM-dev / Documentation

Selection note: Foundation development branch; use a release for a fixed baseline. OpenCFD's separately maintained OpenFOAM distribution has a different upstream and release scheme.

SU2

Multiphysics analysis and design suite centered on CFD and aerodynamic optimization, combining flow solvers, adjoint methods and Python-driven workflows for PDE-constrained engineering design studies.

For: Aerodynamic analysis and adjoint optimization

Established | C++ | NOASSERTION | Last push: 2026-10-24

su2code/SU2 / Documentation

Selection note: Check the configuration and physical-model support for the specific SU2 release; example coverage does not establish accuracy for a new design.

High-order & spectral methods

Discontinuous Galerkin, spectral-element and high-order discretizations.

9 repositories in this topic.

ExaDG

Matrix-free high-order discontinuous Galerkin project whose core incompressible Navier-Stokes solver targets LES and DNS, with additional flow-transport, moving-mesh, fluid-structure and compressible-flow capabilities for research applications.

For: Matrix-free high-order incompressible CFD

Research | C++ | GPL-3.0 | Last push: 2026-09-17

exadg/exadg / Documentation

Selection note: Best aligned with quadrilateral/hexahedral high-order discretizations; installation requires a compatible finite-element and linear-algebra dependency stack.

FLEXI

Discontinuous Galerkin spectral-element framework for hyperbolic-parabolic conservation laws, emphasizing compressible CFD on unstructured hexahedral meshes with parallel execution and a supporting preprocessing and postprocessing suite.

For: High-order compressible-flow simulations

Research | Fortran | GPL-3.0 | Last push: 2026-10-07

flexi-framework/flexi / Documentation

Selection note: Uses a specific high-order mesh and preprocessing workflow; check the required HOPR and HDF5 versions for the chosen release.

HORSES3D

Fortran DG spectral-element multiphysics solver for compressible and incompressible flows, with polynomial adaptivity, turbulence closures and diffuse-interface multiphase formulations on curvilinear hexahedral meshes using hybrid CPU parallelism.

For: Adaptive high-order multiphysics flow studies

Research | Fortran | MIT | Last push: 2026-10-22

horses-framework/horses3d / Documentation

Selection note: The CPU repository and separate GPU project have different supported paths; verify that a desired feature is available in the selected implementation.

MIRGE-Com

Python-driven discontinuous Galerkin simulation application for compressible viscous and inviscid reactive mixtures, using unstructured meshes and accelerator-oriented infrastructure for exascale-enabled scramjet and fluid-flow research.

For: Research on reacting compressible DG solvers

Research | Python | NOASSERTION | Last push: 2026-10-23

illinois-ceesd/mirgecom / Documentation

Selection note: Research application with a specialized array, mesh and code-generation stack; inspect supported chemistry and transport assumptions before selecting it for a study.

Nek5000

Spectral-element solver for incompressible and low-Mach flows, scalar transport and conjugate heat transfer, using high-order tensor-product elements and parallel algorithms suited to turbulence research.

For: High-order incompressible DNS and heat transfer

Established | Fortran | NOASSERTION | Last push: 2026-10-04

Nek5000/Nek5000 / Documentation

Selection note: Classic CPU-oriented Nek implementation; mesh and polynomial resolution must be selected for the intended flow and turbulence regime.

nekRS

High-order spectral-element CFD solver for incompressible and low-Mach flow, with OCCA-based CPU and GPU backends, scalar transport, turbulence models and conjugate heat-transfer capabilities on HPC systems.

For: GPU-oriented spectral-element CFD

Established | C++ | NOASSERTION | Last push: 2026-05-24

Nek5000/nekRS / Documentation

Selection note: Follow the release notes and backend requirements; the project distinguishes stable master from preview development and does not support every desktop platform.

PyFR

Flux-reconstruction CFD platform for high-order scale-resolving flow simulations on unstructured mixed-element meshes, with code-generated computational kernels and multiple CPU and GPU execution backends for heterogeneous systems.

For: High-order scale-resolving engineering flows

Established | Python | NOASSERTION | Last push: 2026-10-21

PyFR/PyFR / Documentation

Selection note: Backend, mesh-element and equation-system support depend on the selected release; compare cost at equivalent accuracy rather than polynomial order alone.

Trixi.jl

Julia framework for high-order conservation-law simulations, combining discontinuous Galerkin and related methods, adaptive tree-based meshes and an extensible equation interface for compressible-flow and numerical-method research.

For: Adaptive high-order conservation-law research

Research | Julia | MIT | Last push: 2026-10-24

trixi-framework/Trixi.jl / Documentation

Selection note: Check equation, mesh and backend compatibility in the documentation; Julia's first-run compilation cost should be separated from steady-state timings.

UCNS3D

Unstructured compressible-flow solver offering high-order finite-volume reconstruction and discontinuous Galerkin methods, mixed-element meshes, turbulence models and MPI/OpenMP execution for aerodynamic and multicomponent-flow research in two or three dimensions.

For: High-order compressible CFD on mixed meshes

Research | Fortran | GPL-3.0 | Last push: 2026-10-23

ucns3d-team/UCNS3D / Documentation

Selection note: Build bounds can be case-specific, and accelerator offload covers selected paths rather than every solver mode; follow the current build instructions.

Turbulence, DNS, LES & HPC

Research solvers, adaptive grids and large-scale flow computing.

12 repositories in this topic.

AFiD

Parallel finite-difference flow solver for Rayleigh-Benard convection and Taylor-Couette turbulence, built around distributed decompositions and FFT-based algorithms for large canonical direct numerical simulations on HPC systems.

For: Canonical convection and rotating-wall DNS

Research | Fortran | NOASSERTION | Last push: 2017-06-15

PhysicsofFluids/AFiD / Documentation

Selection note: Specialized canonical-flow geometry and boundary conditions; check variant-specific documentation and dependencies rather than treating it as a general geometry solver.

CaNS

Parallel incompressible Navier-Stokes solver designed for canonical-flow DNS, using a structured finite-difference discretization and efficient pressure solution with CPU and GPU-oriented implementations for turbulence research at scale.

For: Canonical wall-bounded turbulence DNS

Research | Fortran | MIT | Last push: 2026-10-24

CaNS-World/CaNS / Documentation

Selection note: Targets canonical structured domains; check grid, boundary-condition and accelerator constraints instead of assuming arbitrary industrial geometry support.

Dedalus

Spectral PDE framework with symbolic equation specification and parallel execution, supporting fluid-dynamics research across Cartesian and curved domains rather than restricting users to one fixed equation set.

For: Custom spectral fluid-dynamics models

Established | Python | GPL-3.0 | Last push: 2026-10-24

DedalusProject/dedalus / Documentation

Selection note: Users formulate equations, gauge conditions and boundary constraints; this flexibility is not a substitute for checking mathematical well-posedness and numerical convergence.

FluidSim

Python framework for reproducible fluid-dynamics simulations with extensible solvers, MPI-enabled pseudospectral methods, and integrated output and analysis workflows for turbulence and geophysical-flow research across multiple model equations.

For: Reproducible Python turbulence experiments

Established | Python | NOASSERTION | Last push: 2026-10-23

fluiddyn/fluidsim / Documentation

Selection note: Official publish-only GitHub mirror; canonical development and contributions are on the project's Heptapod Mercurial repository.

IAMR

Parallel adaptive-mesh code for variable-density incompressible Navier-Stokes equations, providing an AMReX-based route to locally refined flow simulations and exploration of block-structured algorithms for research applications.

For: Variable-density incompressible AMR studies

Research | C++ | NOASSERTION | Last push: 2026-10-13

AMReX-Fluids/IAMR / Documentation

Selection note: Variable-density incompressibility is a modeling assumption, not a fully compressible shock-capturing formulation; check the examples and required AMReX revision.

incflo

AMReX-based incompressible Navier-Stokes application for flows in complex geometries, combining block-structured adaptive-grid infrastructure with embedded boundaries and shared hydrodynamic utilities for parallel CPU and GPU research.

For: Adaptive embedded-boundary incompressible CFD

Research | C++ | BSD-3-Clause | Last push: 2026-10-23

AMReX-Fluids/incflo / Documentation

Selection note: Requires compatible AMReX and AMReX-Hydro versions; follow project instructions for supported solvers and boundary conditions rather than assuming general model coverage.

Oceananigans.jl

Julia software for incompressible geophysical fluid dynamics on CPU and GPU hardware, supporting Cartesian and spherical-shell domains, turbulence studies and ocean-oriented models with programmable simulation workflows.

For: GPU geophysical and ocean turbulence

Established | Julia | MIT | Last push: 2026-10-25

CliMA/Oceananigans.jl / Documentation

Selection note: Ocean-oriented incompressible and hydrostatic/nonhydrostatic modeling assumptions differ from general compressible industrial CFD; choose the documented model appropriate to the scales. Its documented ocean and Boussinesq formulations do not cover arbitrary compressible engineering flows.

OpenSBLI

Python code-generation framework translating symbolic compressible-flow equations into parallel high-order finite-difference solvers, with structured curvilinear multiblock meshes and WENO or TENO shock-capturing options for heterogeneous computing architectures.

For: Generated compressible DNS solvers

Research | Python | GPL-3.0 | Last push: 2026-10-12

opensbli/opensbli / Documentation

Selection note: Generated solvers depend on a compatible parallel runtime and compiler stack; changing symbolic equations still requires numerical verification.

Pencil Code

Modular high-order finite-difference code for compressible hydrodynamics with magnetic fields and particles, offering MPI-parallel infrastructure and many sample configurations for turbulence and astrophysical-fluid research on distributed systems.

For: Compressible turbulence and MHD research

Established | Fortran | NOASSERTION | Last push: 2026-10-23

pencil-code/pencil-code / Documentation

Selection note: The project documents its own source-management and contribution workflow; numerical diffusion and shock treatment must be selected for the intended regime.

spectralDNS

Python collection of parallel spectral Navier-Stokes and related solvers, providing an inspectable environment for direct numerical simulation and experiments with spectral discretization and time-integration methods.

For: Python spectral DNS research

Research | Python | LGPL-3.0 | Last push: 2023-08-04

spectralDNS/spectralDNS / Documentation

Selection note: Research-oriented solver collection; verify the maintained solver path and current shenfun/MPI dependency compatibility for the intended example.

WaterLily.jl

Julia solver for unsteady incompressible viscous flow around dynamic immersed bodies, using Cartesian grids, geometric multigrid pressure solves and backend-agnostic CPU or GPU execution in two or three dimensions.

For: Scripted immersed-body flow experiments

Research | Julia | NOASSERTION | Last push: 2026-10-24

WaterLily-jl/WaterLily.jl / Documentation

Selection note: An immersed Cartesian-grid method rather than a body-fitted mesh solver; quantify boundary and grid-resolution effects for the target geometry.

Xcompact3d / Incompact3d

High-order compact finite-difference framework for turbulent-flow DNS and LES, combining Cartesian-grid efficiency, distributed-memory parallelism and immersed-boundary treatments for research on incompressible flows and complex geometries.

For: Structured-grid incompressible DNS and LES

Established | Fortran | BSD-3-Clause | Last push: 2026-01-28

xcompact3d/Incompact3d / Documentation

Selection note: The named repository is the CPU solver; inspect the separate accelerator developments rather than assuming identical GPU capabilities.

Lattice Boltzmann & particle methods

Kinetic, mesoscopic and particle-based fluid formulations.

9 repositories in this topic.

DualSPHysics

Smoothed-particle hydrodynamics solver using C++, CUDA and OpenMP, with engineering workflows focused on free surfaces, wave interactions and large-deformation fluid motion around moving bodies and structures.

For: Free-surface waves and impact problems

Established | C++ | LGPL-2.1 | Last push: 2025-04-16

DualSPHysics/DualSPHysics / Documentation

Selection note: The complete user distribution includes associated tools beyond the source repository; particle resolution, artificial compressibility and boundary treatment require case-specific assessment. Particle resolution, boundary treatment and weak-compressibility settings influence results; GPU builds require a supported CUDA toolchain.

FluidX3D

OpenCL lattice-Boltzmann flow simulator with CPU and GPU execution, integrated visualization, free-surface and thermal extensions, and tools for importing triangulated geometry into a Cartesian lattice for interactive studies.

For: Interactive GPU lattice-Boltzmann studies

Research | C++ | NOASSERTION | Last push: 2026-10-12

ProjectPhysX/FluidX3D / Documentation

Selection note: Source-available with a non-commercial restriction, explicitly not OSI open source. Check its current license before use; the documented solver has no checkpoint/restart support.

GPUSPH

CUDA implementation of weakly compressible smoothed-particle hydrodynamics, providing GPU-oriented infrastructure for free-surface flow simulations and research involving moving boundaries and fluid interactions with engineering geometries.

For: GPU weakly compressible SPH

Research | C++ | NOASSERTION | Last push: 2024-01-28

GPUSPH/gpusph / Documentation

Selection note: CUDA hardware and a compatible compiler stack are required; establish the chosen formulation's resolution and compressibility error for the case.

HemoCell

Cell-resolved suspension framework coupling Palabos lattice-Boltzmann fluid dynamics to deformable Lagrangian cell membranes through immersed boundaries, with dedicated models and parallel data management for blood-flow research.

For: Resolved blood-cell suspension simulations

Research | C++ | NOASSERTION | Last push: 2026-08-01

UvaCsl/HemoCell / Documentation

Selection note: A specialized cellular-flow model, not a clinical prediction tool; inspect cell mechanics, material parameters and case-specific validation requirements.

Lettuce

PyTorch-based lattice-Boltzmann framework for two- and three-dimensional fluid simulations, enabling Python prototyping, GPU execution and experiments connecting kinetic flow solvers with neural networks and automatic differentiation.

For: LBM and machine-learning experiments

Research | Python | MIT | Last push: 2026-10-06

lettucecfd/lettuce / Documentation

Selection note: Check current PyTorch/CUDA compatibility and whether custom kernels support the intended differentiable workflow; default installation targets specific CUDA versions.

NATriuM

High-order off-lattice Boltzmann research solver based on deal.II, supporting unstructured two- and three-dimensional grids through discontinuous-Galerkin or semi-Lagrangian streaming for weakly and fully compressible viscous flows.

For: Off-lattice kinetic-method research reference

Research | C++ | MIT | Last push: 2025-10-02

lettucecfd/NATriuM / Documentation

Selection note: Authors explicitly warn that the code is early-stage and they are unlikely to answer questions; treat it as a research reference with limited support.

PySPH

Python framework for defining particle equations and integrators, generating compiled Cython or OpenCL kernels, and running several SPH formulations with optional shared-memory and distributed parallelism for research simulations.

For: Reproducible SPH method development

Established | Python | NOASSERTION | Last push: 2026-08-23

pypr/pysph / Documentation

Selection note: Different bundled SPH formulations make different assumptions; compare conservation, convergence and boundary behavior for the scheme actually used. A method-development framework: selecting an available formulation does not establish suitability for a particular engineering problem.

Sailfish

GPU lattice-Boltzmann simulation package with a Python-driven setup and CUDA or OpenCL execution, providing examples for exploring kinetic flow models and generated computational kernels on supported accelerators.

For: Reference GPU lattice-Boltzmann implementations

Research | Python | NOASSERTION | Last push: 2022-02-03

sailfish-team/sailfish / Documentation

Selection note: Legacy dependency and GPU-toolchain assumptions need checking before adoption; its historical examples should not be assumed compatible with a current Python environment.

SPHinXsys

C++ multiphysics library built around smoothed-particle hydrodynamics, providing unified discretizations and strongly coupled fluid, solid and multibody models together with examples and a Python interface for engineering studies.

For: Particle-based coupled fluid-solid models

Research | C++ | Apache-2.0 | Last push: 2026-10-23

Xiangyu-Hu/SPHinXsys / Documentation

Selection note: Select the specific formulation and tested example family; a common particle framework does not eliminate resolution or coupling-error studies. Research and engineering library with model-specific assumptions; choose examples matching the intended constitutive laws and flow regime.

Combustion, chemistry & reacting flows

Chemical kinetics, flames, reacting-flow solvers and fire simulation.

11 repositories in this topic.

ABLATE

Develops reacting-flow capabilities around turbulent mixing, fuel entrainment and ablative boundary layers. Offers a research entry point for studying combustion environments associated with hybrid rocket motors.

For: Hybrid-rocket combustion and turbulent entrainment

Research | C++ | BSD-3-Clause | Last push: 2025-05-07

UBCHREST/ablate / Documentation

Selection note: Research-oriented software; use documented examples and confirm the available physical models before planning a new application.

Cantera

Provides thermochemistry, transport properties, reaction mechanisms, reactor networks and one-dimensional flames. A practical foundation for checking chemistry assumptions before embedding them in a reacting-flow solver.

For: Ignition, flame-speed and mechanism studies

Established | C++ | NOASSERTION | Last push: 2026-10-24

Cantera/cantera / Documentation

Selection note: Not a general three-dimensional CFD solver; mechanism suitability must be assessed for the intended conditions.

Fire Dynamics Simulator

Models low-speed fire-driven flows with large-eddy simulation, emphasizing smoke and heat transport. Its documentation includes separate verification and validation material for assessing intended fire-engineering applications.

For: Fire, smoke and building-flow research

Established | Fortran | NOASSERTION | Last push: 2026-10-24

firemodels/fds / Documentation

Selection note: A low-speed fire model rather than a general shock or detonation solver. Safety-critical conclusions require qualified interpretation and appropriate validation.

PeleC

Solves compressible reacting flows using block-structured adaptive meshes and embedded boundaries. Connects detailed chemistry to flow calculations where acoustic waves and compressibility are physically important.

For: Compressible combustion and shock-flame interaction

Established | C++ | NOASSERTION | Last push: 2026-09-28

Pele-Suite/PeleC / Documentation

Selection note: Requires the AMReX, PelePhysics and SUNDIALS stack; mesh, chemistry and temporal resolution need problem-specific assessment.

PeleLMeX

Combines a low-Mach reacting-flow formulation with adaptive refinement and embedded boundaries. Targets flame-turbulence research while avoiding the acoustic time-scale restriction of a fully compressible formulation.

For: Low-Mach turbulent combustion

Research | C++ | BSD-3-Clause | Last push: 2026-10-21

Pele-Suite/PeleLMeX / Documentation

Selection note: The low-Mach formulation is unsuitable when shocks or strong acoustic coupling govern the physics. Use the documented model assumptions.

PelePhysics

Supplies thermodynamics, transport, chemistry generation and additional spray, soot and radiation modules shared by Pele solvers. Useful when researching combustion models rather than duplicating property implementations.

For: Chemistry, spray and soot infrastructure

Established | C++ | NOASSERTION | Last push: 2026-10-22

Pele-Suite/PelePhysics / Documentation

Selection note: A supporting physics library rather than a complete standalone CFD application; module support differs between Pele solvers.

pyJac

Generates analytical Jacobian source code for chemical reaction mechanisms. Helps researchers examine the expensive derivative calculations that arise inside stiff chemical-kinetics integrators and reacting-flow applications.

For: Analytical chemical Jacobian generation

Research | Python | MIT | Last push: 2026-08-10

SLACKHA/pyJac / Documentation

Selection note: An older research tool with version-sensitive dependencies; generated code and mechanism support should be checked in the target environment.

pyMARS

Reduces detailed chemical mechanisms using graph-based methods and sensitivity analysis. Useful for preparing smaller chemistry models when detailed mechanisms make reacting-flow or reactor studies too expensive.

For: Chemical mechanism reduction

Research | Python | MIT | Last push: 2026-06-30

Niemeyer-Research-Group/pyMARS / Documentation

Selection note: Reduction accuracy depends on the selected targets and training conditions; validate the reduced mechanism over the intended operating envelope.

Reaction Mechanism Generator

Constructs chemical kinetic mechanisms using reaction families and thermochemical data. Supports upstream chemistry development for combustion studies before mechanism reduction, reactor assessment or coupling to CFD.

For: Building chemical mechanisms

Established | Python | NOASSERTION | Last push: 2026-10-23

ReactionMechanismGenerator/RMG-Py / Documentation

Selection note: Generated mechanisms are hypotheses requiring assessment against suitable data; this is not a spatial CFD solver.

Spitfire

Builds tabulated chemistry models and integrates chemistry or reaction-diffusion systems. Particularly relevant to nonpremixed flamelet libraries used to reduce the cost of larger fire simulations.

For: Flamelet tables and reaction-diffusion studies

Research | Python | NOASSERTION | Last push: 2026-01-16

sandialabs/Spitfire / Documentation

Selection note: Tables remain specific to their generating assumptions, mechanism and parameter range; extrapolation requires separate assessment.

TChem

Provides chemical source terms, thermodynamic properties, Jacobians and canonical reactor formulations for complex kinetic mechanisms. Useful for integrating and studying the stiff chemistry component of reacting simulations.

For: Chemical kinetics kernels and reactor models

Established | Jupyter Notebook | BSD-2-Clause | Last push: 2025-08-29

sandialabs/TChem / Documentation

Selection note: Distinguish this kinetics toolkit from the separate TChem-atm atmospheric-chemistry project; it does not supply a complete spatial flow solver.

Multiphase flow, droplets & sprays

Interfaces, dispersed phases, droplets and free-surface simulation.

5 repositories in this topic.

CFDEM coupling PUBLIC

Couples OpenFOAM flow calculations with discrete particle dynamics through selectable force and data-exchange models. Useful for studying fluidized beds, particulate transport and fluid-particle interaction workflows.

For: CFD-DEM particulate flow

Established | C | GPL-3.0 | Last push: 2025-07-03

CFDEMproject/CFDEMcoupling-PUBLIC / Documentation

Selection note: Public and commercial product capabilities differ. Verify the documented OpenFOAM and particle-solver versions before installation.

JAX-Fluids

Implements differentiable compressible flow simulation with high-order reconstruction, level-set and diffuse-interface two-phase methods. Enables research connecting numerical fluid dynamics, automatic differentiation and accelerator-based computation.

For: Differentiable compressible multiphase research

Research | Python | NOASSERTION | Last push: 2026-08-28

tumaer/JAXFLUIDS / Documentation

Selection note: Differentiating a discrete solver does not automatically yield reliable gradients through shocks or interface events; examine discretization and sensitivity assumptions.

Mantaflow

Combines a fluid-simulation core with Python scene definitions and an extensible plugin system. Supports liquid, smoke and learning experiments aimed at graphics-oriented fluid algorithm research.

For: Liquid and smoke algorithm prototyping

Research | C++ | NOASSERTION | Last push: 2022-10-26

thunil/mantaflow / Documentation

Selection note: Designed primarily for computer-graphics research; attractive visual output alone does not establish engineering accuracy.

MFC

Solves compressible multiphase flows with interface models, bubble dynamics, phase-change options and particle tracking. Useful for investigating shock-interface interaction and other strongly compressible multiphase phenomena.

For: Shock-bubble interaction and compressible mixtures

Established | Fortran | MIT | Last push: 2026-10-24

MFlowCode/MFC / Documentation

Selection note: Different mixture, interface and bubble models make different assumptions; select the model appropriate to the physical scale and regime.

SPlisHSPlasH

Implements several incompressible SPH pressure solvers alongside viscosity, surface tension and rigid or deformable body coupling. Particularly useful for comparing particle-fluid simulation methods and visual experiments.

For: SPH incompressibility and coupled particle scenes

Established | C++ | MIT | Last push: 2026-10-16

InteractiveComputerGraphics/SPlisHSPlasH / Documentation

Selection note: Originates in physically based graphics research; engineering use needs independent dimensional, resolution and validation checks.

Coupled physics & specialist applications

Fluid-structure interaction, porous flow, biomedical flow and design.

10 repositories in this topic.

DAFoam

Adds discrete-adjoint capabilities to OpenFOAM-based multiphysics analysis with a Python interface for optimization. Supports aerodynamic, thermal and coupled design studies with many design variables.

For: Adjoint aerodynamic and multidisciplinary optimization

Established | C | NOASSERTION | Last push: 2026-10-19

mdolab/dafoam / Documentation

Selection note: Requires compatible OpenFOAM and optimization dependencies; convergence of the primal and adjoint problems matters for usable design gradients. Use the documented solver and dependency versions; verify adjoint gradients for the chosen objective and case.

DuMux

Models porous-medium flow, transport and deformation with finite-volume and control-volume finite-element methods. Includes coupled multidomain formulations for applications that cross free-flow and porous regions.

For: Porous-media and multidomain transport

Established | C++ | GPL-3.0 | Last push: 2026-10-24

dumux/dumux / Documentation

Selection note: The GitHub repository is an official mirror; primary development and issue tracking are hosted by the Stuttgart GitLab project.

IBAMR

Combines immersed-boundary methods with Cartesian adaptive mesh refinement and distributed-memory parallelism. Useful for deformable structures interacting with fluids while retaining separate Eulerian and Lagrangian descriptions.

For: Immersed-boundary fluid-structure interaction

Established | C++ | NOASSERTION | Last push: 2026-10-25

IBAMR/IBAMR / Documentation

Selection note: Requires a substantial numerical-library stack; discretization and coupling choices must match the stiffness, resolution and boundary conditions of the problem.

MRST

Provides a MATLAB toolbox for flow, transport and mechanics in porous media. Includes modular reservoir and carbon-storage workflows that support numerical experimentation and model development.

For: Reservoir simulation and carbon storage

Established | MATLAB | GPL-3.0 | Last push: 2026-10-20

SINTEF-AppliedCompSci/MRST / Documentation

Selection note: Upstream recommends packaged releases for most users; compatibility with GNU Octave varies by module and should not be assumed.

oomph-lib

Offers an object-oriented finite-element library with extensive multiphysics demonstration problems. Useful for building fluid-mechanics and fluid-structure models when researchers need control of the governing formulation.

For: Finite-element multiphysics method development

Established | C++ | NOASSERTION | Last push: 2026-10-22

oomph-lib/oomph-lib / Documentation

Selection note: A programmable numerical library rather than a turnkey engineering interface; model formulation and solver choices remain the user's responsibility.

OpenPNM

Represents porous materials as pore networks and solves transport across them. Useful when investigating effective flow and transport properties without directly resolving every pore-scale fluid volume.

For: Pore-network transport modeling

Established | Python | MIT | Last push: 2026-07-21

PMEAL/OpenPNM / Documentation

Selection note: Network representations simplify pore geometry and physics; they are not interchangeable with fully resolved pore-scale Navier-Stokes simulations.

OPM Flow

Implements reservoir flow simulation with automatic differentiation and established reservoir input formats. Useful for multiphase subsurface studies that need well models and reservoir-scale fluid transport.

For: Reservoir-scale multiphase simulation

Established | C++ | GPL-3.0 | Last push: 2026-10-24

OPM/opm-simulators / Documentation

Selection note: Reservoir models operate under subsurface-flow assumptions and should not be substituted directly for free-fluid CFD.

preCICE

Connects existing simulation programs through partitioned coupling, interface data exchange and mapping. Provides a common route to fluid-structure interaction, conjugate heat transfer and other coupled workflows.

For: Coupling independent physics solvers

Established | C++ | LGPL-3.0 | Last push: 2026-10-02

precice/precice / Documentation

Selection note: A coupling library, not a CFD solver. Stable coupling still depends on compatible adapters, interface conditions and convergence settings.

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solids4foam

Extends the OpenFOAM ecosystem with solid-mechanics and fluid-solid interaction models. Provides tutorials for comparing partitioned coupling approaches and examining deformation-driven changes to a flow domain.

For: Finite-volume fluid-solid interaction

Established | C | GPL-3.0 | Last push: 2026-10-25

solids4foam/solids4foam / Documentation

Selection note: Compatibility is OpenFOAM-version dependent; verify the supported fork and version rather than mixing tutorial generations.

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svMultiPhysics

Provides coupled finite-element models for fluids, solids, diffusion and electrophysiology in cardiovascular research. Supports investigations where blood flow and deforming biological tissue must be modeled together.

For: Cardiovascular multiphysics and fluid-structure research

Research | C++ | BSD-3-Clause | Last push: 2026-10-23

SimVascular/svMultiPhysics / Documentation

Selection note: Formerly svFSIplus; research software does not itself establish clinical suitability. Patient-specific predictions require separate validation and appropriate expertise.

Ocean, atmosphere, wind & environmental flow

Geophysical, coastal, hydrological and wind-energy modelling.

16 repositories in this topic.

ADCIRC

Models time-dependent free-surface circulation and transport on unstructured finite-element grids. Common application areas include storm surge, tides, wind-driven circulation and coastal flooding studies.

For: Storm surge and coastal circulation

Established | Fortran | NOASSERTION | Last push: 2026-10-20

adcirc/adcirc / Documentation

Selection note: Requires suitable meshes, forcing and calibration for the site; official documentation is being consolidated across several maintained sources.

ANUGA

Solves shallow-water equations for floods, tsunamis and storm surges using finite-volume methods. Provides Python workflows and examples for combining terrain, boundary conditions and inundation studies.

For: Flood and tsunami inundation

Established | Python | NOASSERTION | Last push: 2026-10-24

anuga-community/anuga_core / Documentation

Selection note: Depth-averaged shallow-water assumptions do not resolve detailed vertical flow structure; terrain and boundary-data quality strongly affect inundation results.

CM1

Supports idealized atmospheric experiments with a nonhydrostatic numerical model. Useful for studying convection, clouds and organized storms while controlling the background state and experimental forcing.

For: Idealized convection and storm research

Established | Fortran | MIT | Last push: 2026-08-05

NCAR/CM1 / Documentation

Selection note: Primarily an idealized research model; configuring a real-time operational forecast is a different task from running its standard experiments.

Delft3D

Collects Delft3D structured and flexible-mesh simulation engines for hydrodynamics, waves, morphology and water quality. Useful for integrated coastal, estuarine, river and urban water studies.

For: Integrated hydrodynamics and morphodynamics

Established | Fortran | NOASSERTION | Last push: 2026-10-24

Deltares/Delft3D / Documentation

Selection note: The source engines and packaged graphical interfaces have different delivery arrangements; component licensing and build requirements should be checked individually.

ERF

Builds atmospheric-flow modeling on AMReX with adaptive meshes and a documented theory and examples. Relevant to energy-related atmospheric research and simulations requiring local resolution refinement.

For: Adaptive atmospheric and energy-flow studies

Research | C++ | NOASSERTION | Last push: 2026-10-25

erf-model/ERF / Documentation

Selection note: Check the supported atmospheric models and boundary treatments for the intended application; the solver remains an evolving research code.

FUNWAVE-TVD

Models dispersive nearshore waves with a Boussinesq-type formulation and shock-capturing treatment. Useful for wave transformation, coastal wave generation and studying processes beyond nondispersive shallow-water models.

For: Dispersive coastal-wave transformation

Research | Fortran | NOASSERTION | Last push: 2025-01-24

fengyanshi/FUNWAVE-TVD / Documentation

Selection note: Upstream labels the default branch beta and directs users to released versions. Confirm licensing and the release appropriate to your study.

GeoClaw

Extends Clawpack for geophysical waves and flows, including tsunami propagation and inundation. Uses adaptive refinement to concentrate resolution where evolving waves and coastal interactions require it.

For: Tsunami propagation and inundation

Established | Fortran | BSD-3-Clause | Last push: 2026-10-24

clawpack/geoclaw / Documentation

Selection note: Depth-averaged flow assumptions and topographic resolution constrain the phenomena represented; assess the model for the intended hazard application.

Kynema-SGF (formerly AMR-Wind)

Uses an adaptive structured-grid incompressible formulation for atmospheric boundary layers and wind-farm wakes. Can serve as the background flow solver in coupled blade-resolved wind-energy simulations.

For: Wind-farm wakes and atmospheric boundary layers

Established | C++ | NOASSERTION | Last push: 2026-10-20

kynema/kynema-sgf / Documentation

Selection note: Renamed from AMR-Wind in 2026; actuator models and blade-resolved overset configurations represent different levels of physical detail.

Kynema-UGF (formerly Nalu-Wind)

Provides an unstructured incompressible flow solver used in the Kynema wind-energy ecosystem. Useful for geometry-resolved turbine and energy-system flow calculations alongside structured background simulations.

For: Geometry-resolved wind and energy-system flows

Established | C | NOASSERTION | Last push: 2026-10-18

kynema/kynema-ugf / Documentation

Selection note: Renamed from Nalu-Wind in 2026; coupled overset studies require additional ecosystem components and matching configuration.

MITgcm

Models ocean and atmospheric circulation with a configurable numerical core and extensive verification examples. Useful across geophysical scales, including nonhydrostatic ocean processes and large-scale circulation.

For: Ocean circulation and geophysical fluid dynamics

Established | Fortran | MIT | Last push: 2026-10-23

MITgcm/MITgcm / Documentation

Selection note: Configuration, forcing and parameterizations determine the applicable scale; reproducing an experiment requires more than selecting a solver executable.

MOM6

Solves ocean primitive equations with flexible vertical coordinates and options for coupled Earth-system configurations. Useful for ocean circulation studies and research into ocean-model numerics and parameterizations.

For: Ocean circulation and Earth-system modeling

Established | Fortran | Apache-2.0 | Last push: 2026-10-18

NOAA-GFDL/MOM6 / Documentation

Selection note: Designed for ocean primitive-equation modeling rather than general free-surface engineering CFD; regional setups need appropriate forcing and boundary data.

MPAS-Model

Provides a shared unstructured-mesh framework with atmosphere, ocean and other geophysical model cores. Useful for multiscale circulation studies where mesh design and regional refinement matter.

For: Variable-resolution geophysical modeling

Established | Fortran | NOASSERTION | Last push: 2026-10-13

MPAS-Dev/MPAS-Model / Documentation

Selection note: Individual cores have separate build instructions, capabilities and data requirements; the shared repository is not a single universal model.

Proteus

Separates continuum physical models from numerical methods in a Python-led toolkit. Includes multiphase Navier-Stokes, shallow-water and porous-flow formulations relevant to hydraulic and coastal research.

For: Hydraulic multiphase and numerical-method research

Established | Jupyter Notebook | MIT | Last push: 2026-09-04

erdc/proteus / Documentation

Selection note: Broad method availability does not make every formulation suitable for every problem; external compiled dependencies are part of the installation.

SCHISM

Uses unstructured grids for three-dimensional baroclinic circulation across connected river, estuary, shelf and ocean scales. Useful when one study spans several interacting hydrodynamic environments.

For: Cross-scale river-estuary-ocean circulation

Established | LLVM | Apache-2.0 | Last push: 2026-10-22

schism-dev/schism / Documentation

Selection note: Choose supported modules and document forcing and vertical-grid assumptions; development-branch flags can change between model versions.

Thetis

Builds coastal and estuarine flow models on Firedrake finite elements. Includes shallow-water, tracer, wetting-and-drying and adjoint capabilities for studying circulation and inverse problems.

For: Coastal circulation and adjoint studies

Established | Python | NOASSERTION | Last push: 2026-10-22

thetisproject/thetis / Documentation

Selection note: Development versions may require matching Firedrake development branches; shallow-water assumptions must fit the intended physical scales.

WRF

Provides the Weather Research and Forecasting modeling system for atmospheric research and numerical weather prediction. Useful when CFD studies need meteorological context or mesoscale flow conditions.

For: Weather and mesoscale atmospheric modeling

Established | Fortran | NOASSERTION | Last push: 2026-10-22

wrf-model/WRF / Documentation

Selection note: Meteorological predictions depend on initialization, forcing and physical parameterizations; WRF is not a replacement for geometry-resolved local engineering CFD.

Geometry, CAD & mesh generation

Geometry preparation, mesh generation, adaptation and quality tools.

12 repositories in this topic.

CadQuery

Python-based parametric CAD scripting built on Open CASCADE. Parameterized channels, nozzles and mechanical components can be generated consistently for geometry sweeps, automated studies and downstream mesh preparation.

For: Scripted geometry sweeps

Established | Python | NOASSERTION | Last push: 2026-10-24

CadQuery/cadquery / Documentation

Selection note: Provides geometry construction rather than a CFD mesher or solver.

CGAL

Computational geometry algorithms for triangulation, surface processing and mesh generation. Supplies robust geometric building blocks for custom CFD preprocessing tools rather than a single ready-made flow-analysis workflow.

For: Geometry algorithms and mesh infrastructure

Established | C++ | NOASSERTION | Last push: 2026-10-23

CGAL/cgal / Documentation

Selection note: License terms differ by CGAL package; check the specific component before integration.

FreeCAD

Parametric solid modelling for engineering geometry, with editable design history and interoperable CAD workflows. Useful for preparing and revising fluid-domain boundaries before meshing rather than solving flow.

For: Parametric engineering geometry

Established | C++ | LGPL-2.1 | Last push: 2026-10-24

FreeCAD/FreeCAD / Documentation

Selection note: A CAD model is not automatically a watertight or appropriately resolved CFD domain.

IDWarp

Deforms volume mesh nodes in response to surface perturbations using an inverse-distance approach. Useful for reusing a baseline CFD mesh during aerodynamic shape optimization and sensitivity calculations.

For: Volume-mesh deformation

Established | Fortran | NOASSERTION | Last push: 2026-09-06

mdolab/idwarp / Documentation

Selection note: Large surface changes can still produce poor-quality or invalid cells; monitor mesh quality.

meshio

Reads and writes many mesh formats through Python and a command-line interface. Helps move geometry, cells and associated fields between meshing tools, numerical solvers and visualization software.

For: Mesh-format conversion

Established | Python | MIT | Last push: 2024-07-23

nschloe/meshio / Documentation

Selection note: Not every format preserves every field, boundary convention or high-order element representation.

Netgen

Automatic tetrahedral mesh generation from constructive geometry, STL surfaces and CAD geometry. Python scripting, refinement and optimization make it useful for reproducible finite-element flow preprocessing.

For: Scriptable tetrahedral meshes

Established | C++ | LGPL-2.1 | Last push: 2026-10-24

NGSolve/netgen / Documentation

Selection note: Mesh topology, boundary tagging and wall-resolution requirements still need solver-specific checking.

pygalmesh

Python interface to CGAL's mesh-generation capabilities, including primitive and implicit domains. Useful for constructing tetrahedral meshes from mathematical geometries and exporting them for numerical PDE experiments.

For: Implicit-domain tetrahedral meshing

Established | C++ | GPL-3.0 | Last push: 2024-07-22

meshpro/pygalmesh / Documentation

Selection note: Sharp intersections and detailed meshing controls need care; this is not a CAD repair application.

pyGeo

Geometry parameterization and geometric constraints for multidisciplinary optimization. Free-form deformation, CAD interfaces and design-variable handling connect aerodynamic shape changes to repeatable CFD and coupled engineering studies.

For: Shape parameterization for CFD optimization

Established | Python | Apache-2.0 | Last push: 2026-08-24

mdolab/pygeo / Documentation

Selection note: Individual CAD interfaces introduce their own installation and licensing requirements.

pygmsh

Python geometry helpers around Gmsh reduce repetitive low-level entity bookkeeping. Useful for creating parameterized two- and three-dimensional domains and exporting meshes into automated numerical experiments.

For: Python-driven Gmsh geometry

Established | Python | GPL-3.0 | Last push: 2023-10-04

nschloe/pygmsh / Documentation

Selection note: Requires Gmsh; native Gmsh features and version compatibility should be checked for advanced workflows.

pyHyp

Extrudes structured surface meshes into volume meshes using hyperbolic marching. Designed for the preprocessing stage of aerodynamic optimization workflows where body-fitted structured grids and controlled extrusion are useful.

For: Structured aerodynamic volume meshes

Established | Fortran | NOASSERTION | Last push: 2026-09-05

mdolab/pyhyp / Documentation

Selection note: Needs a suitable structured surface mesh; it is not an arbitrary-geometry unstructured mesher.

SALOME SMESH

SALOME's mesh module creates, edits, imports and exports computational meshes, with grouping, quality controls and geometry integration. Supports preprocessing pipelines needing interactive inspection alongside structured automation.

For: Interactive mesh preparation and quality checks

Established | C++ | LGPL-2.1 | Last push: 2026-09-14

SalomePlatform/smesh / Documentation

Selection note: Building the module requires additional SALOME components; the repository is not a standalone application installer.

trimesh

Python tools for loading, inspecting and manipulating triangular surfaces. Helpful for checking imported STL geometry, connectivity and surface properties before committing to expensive volume meshing or flow calculations.

For: STL surface inspection

Established | Python | MIT | Last push: 2026-10-24

mikedh/trimesh / Documentation

Selection note: Surface processing does not establish CAD fidelity or the adequacy of a CFD boundary-layer mesh.

Visualization, post-processing & I/O

Inspect, process, exchange and communicate simulation data.

10 repositories in this topic.

ADIOS2

Scalable scientific input/output framework transporting self-describing variables through files, networks and memory. Useful for large CFD output pipelines, streaming analysis and reducing coupling between simulation and data consumers.

For: HPC output and streaming pipelines

Established | C++ | Apache-2.0 | Last push: 2026-10-24

ornladios/ADIOS2 / Documentation

Selection note: Requires integration into the application or an existing compatible I/O path.

CGNS

Defines and implements a standard for storing CFD meshes, flow solutions and related metadata. Useful when interoperability and consistent exchange between preprocessing, simulation and postprocessing tools matter.

For: Standardized CFD data exchange

Established | C | NOASSERTION | Last push: 2026-10-19

CGNS/CGNS / Documentation

Selection note: A data model and library, not a solver; interoperable use still depends on supported CGNS conventions.

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netCDF4-Python

Python and NumPy interface to the netCDF library. Useful for reading, writing and inspecting array-based simulation datasets, especially geophysical and atmospheric fluid workflows using netCDF conventions.

For: Geophysical CFD array I/O

Established | Python | MIT | Last push: 2026-10-23

Unidata/netcdf4-python / Documentation

Selection note: Respect netCDF thread-safety limitations and preserve units, coordinates and metadata conventions.

ParaView

Scientific visualization application built on VTK, supporting large-data analysis and programmable pipelines. Widely useful for CFD field inspection, slices, streamlines, time histories and reproducible postprocessing workflows.

For: General CFD postprocessing

Established | C++ | BSD-3-Clause | Last push: 2026-10-24

Kitware/ParaView / Documentation

Selection note: Official GitHub mirror; upstream development and issue reporting take place on Kitware GitLab.

PyVista

NumPy-friendly Python interface to VTK datasets, filters and plotting. Supports scripted extraction, slicing and interactive display of CFD meshes and fields in notebooks, applications or automated reporting.

For: Python mesh and field analysis

Established | Python | MIT | Last push: 2026-10-25

pyvista/pyvista / Documentation

Selection note: Headless rendering and remote notebook use may require additional graphics configuration.

trame

Python framework for interactive scientific web applications, integrating VTK and ParaView rendering. Useful for turning CFD analysis pipelines into browser-accessible viewers with local or remote visualization.

For: Browser-based CFD viewers

Established | Python | NOASSERTION | Last push: 2026-09-28

Kitware/trame / Documentation

Selection note: Application deployment, authentication and resource limits need separate design for public-facing services.

vedo

Python scientific visualization built on VTK and NumPy, with mesh manipulation and analysis helpers. Useful for compact exploratory scripts, interactive inspection and presentation of surfaces or volumetric flow data.

For: Compact interactive visualization scripts

Established | Python | MIT | Last push: 2026-08-04

marcomusy/vedo / Documentation

Selection note: Check supported readers and field association conventions against the output format of your solver.

VisIt

Scientific visualization and analysis application for mesh-based simulation data. Useful for exploring large CFD datasets, applying derived quantities and building repeatable analysis routines across supported simulation formats.

For: Large mesh-based simulation analysis

Established | C | BSD-3-Clause | Last push: 2026-10-22

visit-dav/visit / Documentation

Selection note: Solver output must match a supported reader or be converted to an appropriate exchange format.

VTK

Visualization and data-processing toolkit for building scientific applications. Provides meshes, filters, rendering and file readers that underpin custom CFD postprocessing tools and many established visualization applications.

For: Custom scientific visualization software

Established | C++ | NOASSERTION | Last push: 2026-10-25

Kitware/VTK / Documentation

Selection note: A development toolkit rather than a complete end-user CFD application; GitHub is an official mirror.

yt

Python analysis and visualization for volumetric simulation datasets, including adaptive meshes. Useful for derived fields, spatial selections, profiles and volume rendering in computational astrophysics and related fluid simulations.

For: AMR and astrophysical-fluid data analysis

Established | Python | NOASSERTION | Last push: 2026-10-24

yt-project/yt / Documentation

Selection note: Strongest fit is supported volumetric simulation formats; confirm a frontend exists for your solver.

Scientific ML, differentiable CFD & ROM

Physics-informed learning, operators, differentiable models and reduced-order methods.

12 repositories in this topic.

DAPPER

Experimental data-assimilation framework with ensemble and variational methods, diagnostics and model examples. Useful for learning state estimation and testing assimilation ideas on geophysical-fluid prototypes before solver-specific integration.

For: Data-assimilation method comparisons

Research | Python | NOASSERTION | Last push: 2026-10-12

nansencenter/DAPPER / Documentation

Selection note: Designed for teaching and fundamental experiments; coupling production CFD requires additional implementation.

DeepXDE

Scientific machine-learning library supporting physics-informed networks and neural operators. Includes differential-equation workflows useful for fluid inverse problems, parameter estimation and exploratory surrogate studies with explicit boundary and initial conditions.

For: PINN and inverse-flow experiments

Established | Python | LGPL-2.1 | Last push: 2026-08-18

lululxvi/deepxde / Documentation

Selection note: PINN training convergence is problem dependent; verify predictions against trusted numerical or experimental references.

EZyRB

Data-driven reduced-basis modelling from parameterized numerical snapshots. Useful for constructing nonintrusive CFD surrogates when full-order results are available but modifying the original governing-equation implementation is impractical.

For: Nonintrusive parametric CFD surrogates

Research | Jupyter Notebook | NOASSERTION | Last push: 2026-05-25

mathLab/EZyRB / Documentation

Selection note: Prediction quality depends on snapshot coverage; extrapolation beyond the training parameter range needs testing.

ITHACA-FV

OpenFOAM-based reduced-order-modelling framework with methods for developing compact finite-volume models. Useful for experimenting with projection, parameterized flow snapshots and reduced solvers close to existing OpenFOAM workflows.

For: Reduced models built around OpenFOAM

Research | C++ | LGPL-3.0 | Last push: 2025-05-13

mathLab/ITHACA-FV / Documentation

Selection note: Check the supported OpenFOAM release and rebuild dependencies for the selected branch.

NeuralOperator

Official neural-operator library containing Fourier Neural Operators and related architectures. Useful for learning mappings between PDE inputs and solution fields, including fluid and transport surrogate research.

For: Fourier neural-operator baselines

Research | Python | MIT | Last push: 2026-10-17

neuraloperator/neuraloperator / Documentation

Selection note: Resolution transfer does not guarantee accuracy under new geometries, regimes or boundary conditions.

NVIDIA PhysicsNeMo

PyTorch-based physics machine-learning framework with reusable models and training recipes. Aerodynamics examples make it relevant to surface and volume surrogates, scalable training and engineering field prediction.

For: Scalable engineering surrogates

Established | Python | Apache-2.0 | Last push: 2026-10-25

NVIDIA/physicsnemo / Documentation

Selection note: GPU requirements, model prerequisites and separate dataset terms vary across examples.

PDEArena

Training and evaluation framework for neural PDE modelling, with architecture collections and fluid datasets. Useful for controlled comparisons of learned Navier-Stokes and shallow-water evolution models across spatial and temporal scales.

For: Fluid surrogate benchmark experiments

Research | Python | MIT | Last push: 2024-09-14

pdearena/pdearena / Documentation

Selection note: Research benchmarks and trained models do not establish performance on industrial geometries or operating regimes.

PhiFlow

Differentiable simulation toolkit integrated with NumPy and machine-learning frameworks. Provides fluid examples and automatic differentiation pathways useful for inverse problems, learned corrections and optimization through simulations.

For: Differentiable fluid-learning experiments

Research | Python | MIT | Last push: 2026-07-16

tum-pbs/PhiFlow / Documentation

Selection note: Choose the numerical model and discretization for the intended physics; differentiability alone does not establish fidelity.

PyDMD

Dynamic mode decomposition tools for extracting coherent spatial and temporal structures from snapshot data. Useful for analyzing CFD transients, building data-driven reduced models and comparing modal representations.

For: Flow-snapshot modal analysis

Established | Python | MIT | Last push: 2026-10-03

PyDMD/PyDMD / Documentation

Selection note: Mode estimates depend on sampling, noise and preprocessing; the fitted dynamics need independent validation.

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pyMOR

Model-order-reduction library using abstract interfaces to connect numerical algorithms and PDE backends. Supports reduced bases, POD and system-theoretic approaches for constructing compact parameterized flow or transport models.

For: Projection-based reduced models

Established | Python | NOASSERTION | Last push: 2026-10-21

pymor/pymor / Documentation

Selection note: Connecting a new CFD solver requires compatible operators, data access and suitable reduction assumptions.

PySINDy

Sparse system-identification tools for discovering compact dynamical models from data. Useful for fluid reduced coordinates and PDE-discovery experiments where interpretable model structure is preferable to a black-box surrogate.

For: Data-driven fluid dynamics identification

Research | Python | NOASSERTION | Last push: 2026-10-15

dynamicslab/pysindy / Documentation

Selection note: Derivative estimation, noise and candidate-library choices strongly affect discovered equations.

Transolver

Research implementation of transformer-based PDE modelling on general geometries. Includes airfoil and car-flow design experiments, making it useful for studying geometry-aware surrogates alongside established neural-operator baselines.

For: Geometry-aware aerodynamic surrogates

Research | Python | MIT | Last push: 2026-02-26

thuml/Transolver / Documentation

Selection note: Published benchmark claims belong to the original study; reproduce the selected task before relying on generalization.

Verification, benchmarks & learning

Teaching material, reference cases, datasets and numerical checks.

11 repositories in this topic.

CFD Python

A notebook sequence teaching CFD through incremental PDE implementations, culminating in Navier-Stokes exercises. Useful for understanding discretization, numerical diffusion and convergence before adopting larger solver frameworks.

For: First-principles CFD learning

Established | Jupyter Notebook | NOASSERTION | Last push: 2024-03-19

barbagroup/CFDPython / Documentation

Selection note: Educational discretizations and small examples are not production CFD solvers.

CFDBench

Fluid-dynamics surrogate benchmark with cavity, tube, dam and cylinder cases across boundary conditions, geometries and physical properties. Includes datasets and baseline implementations for studying inference-time generalization.

For: Parameterized CFD surrogate comparisons

Research | Python | NOASSERTION | Last push: 2025-10-25

luo-yining/CFDBench / Documentation

Selection note: Original data generation uses commercial ICEM/Fluent; using released data has separate download and storage requirements.

FEniCSx Tutorial

Executable finite-element tutorial covering DOLFINx fundamentals and Navier-Stokes examples. Useful for deriving weak forms, applying boundary conditions and exploring transparent flow implementations with associated explanations and parallel execution.

For: Finite-element flow implementation learning

Established | Jupyter Notebook | NOASSERTION | Last push: 2026-10-06

jorgensd/dolfinx-tutorial / Documentation

Selection note: Follow the tutorial's supported DOLFINx version; examples are not a universal production flow solver.

Giverny / JHTDB access

Python and MATLAB access tools for querying the Johns Hopkins Turbulence Database. Enables sampled DNS fields and cutouts for turbulence analysis, numerical comparisons and data-driven research without downloading entire simulations.

For: Reference DNS turbulence queries

Established | Python | Apache-2.0 | Last push: 2026-10-23

sciserver/giverny / Documentation

Selection note: Access tokens, query-size limits and available variables depend on the service and chosen dataset.

MASA

Library of manufactured analytical solutions and source terms for PDE software verification. Includes Euler, Navier-Stokes and turbulence-model formulations useful for checking implementation errors and observed convergence orders.

For: Method-of-manufactured-solutions verification

Established | C++ | NOASSERTION | Last push: 2025-10-28

manufactured-solutions/MASA / Documentation

Selection note: Manufactured solutions verify equation implementation; they do not validate physical modelling against experiments.

NekExamples

Official Nek5000 examples with case descriptions and required simulation files. Includes canonical and turbulent-flow setups useful for learning spectral-element workflows, inspecting boundary conditions and checking an installation.

For: Nek5000 example and verification cases

Established | Python | NOASSERTION | Last push: 2021-05-25

Nek5000/NekExamples / Documentation

Selection note: Examples require Nek5000 and case-specific preparation; confirm mesh resolution and expected reference quantities.

PDEBench

Scientific machine-learning benchmark combining PDE datasets, data-generation code and baseline models. Fluid cases and varied initial, boundary and physical parameters support repeatable comparisons of learned flow prediction methods.

For: Standardized fluid-surrogate evaluation

Research | Python | NOASSERTION | Last push: 2026-03-30

pdebench/PDEBench / Documentation

Selection note: Datasets are hosted separately and can be large; check documented environment and data-path requirements.

preCICE Tutorials

Worked coupled-simulation cases using preCICE with real solvers, including fluid-structure and thermal applications. Useful for understanding participant configuration, adapter interfaces and reproducible multiphysics setup rather than isolated equations.

For: Coupled CFD learning cases

Established | C | LGPL-3.0 | Last push: 2026-10-24

precice/tutorials / Documentation

Selection note: Most tutorials need multiple solver installations and adapters with compatible versions.

Riemann Problems and Jupyter Solutions

Interactive notebook book covering Riemann problems for hyperbolic equations, including acoustics, shallow water and Euler flow. Useful for understanding wave structure and comparing exact or approximate flux solvers.

For: Hyperbolic-flow numerical methods

Established | HTML | BSD-3-Clause | Last push: 2025-11-30

clawpack/riemann_book / Documentation

Selection note: Text and executable code have separate license files; consult both when reusing material.

Riemann-Solvers

Compact implementations accompanying classical Riemann-solver methods for advection, Burgers and Euler equations. Useful for comparing schemes and developing intuition about shocks, rarefactions and numerical fluxes.

For: Hands-on flux-solver comparison

Research | C++ | NOASSERTION | Last push: 2019-03-28

cangyu/Riemann-Solvers / Documentation

Selection note: No repository-level license was apparent in the inspected listing; inspect permissions before copying code.

SU2 TestCases

Official collection of SU2 test inputs spanning flow, adjoints, conjugate heat transfer and other capabilities. Useful for reproducing regression cases and finding solver-specific examples matched to SU2 development.

For: SU2 regression and setup examples

Established | GLSL | LGPL-2.1 | Last push: 2026-10-23

su2code/TestCases / Documentation

Selection note: Match the case branch or revision to the SU2 solver version; a regression pass is not physical validation.

Automation, optimization & uncertainty

Workflow orchestration, adjoints, sensitivity analysis and uncertainty quantification.

11 repositories in this topic.

AeroSandbox

Python aircraft design and optimization environment using automatic differentiation and composable engineering analyses. Useful for rapid aerodynamic design exploration and multidisciplinary studies before committing to higher-fidelity CFD evaluations.

For: Aircraft design and aerodynamic screening

Established | Jupyter Notebook | MIT | Last push: 2026-10-20

peterdsharpe/AeroSandbox / Documentation

Selection note: Many included methods are reduced-fidelity engineering models; do not equate all analyses with Navier-Stokes CFD.

Chaospy

Numerical toolbox for polynomial chaos, sampling and quadrature. Useful for propagating uncertain flow parameters through CFD surrogates or moderately priced solvers and building nonintrusive uncertainty-expansion experiments.

For: Polynomial-chaos CFD uncertainty studies

Established | Python | MIT | Last push: 2025-09-04

jonathf/chaospy / Documentation

Selection note: High-dimensional or discontinuous response surfaces may require alternatives to a simple polynomial expansion.

Dakota

Design exploration framework combining optimization, parameter estimation, sensitivity analysis and uncertainty quantification. Useful for managing repeated CFD evaluations and propagating uncertain inputs through external engineering simulation models.

For: External-solver design exploration and UQ

Established | C++ | NOASSERTION | Last push: 2026-10-24

snl-dakota/dakota / Documentation

Selection note: Requires a working simulation interface and can demand many expensive flow evaluations.

foamlib

Python interface for OpenFOAM files, cases and execution. Useful for reproducible parameter studies, programmatic field access and reducing repetitive case-management steps around an existing OpenFOAM installation.

For: Automated OpenFOAM case management

Emerging | Python | GPL-3.0 | Last push: 2026-10-23

gerlero/foamlib / Documentation

Selection note: OpenFOAM must be installed separately; confirm support for the selected distribution and file types.

MPhys

Standardizes high-fidelity multiphysics problems within OpenMDAO using modular conventions and solver integrations. Useful for assembling CFD-structures or aerothermal studies while preserving coupling and derivative interfaces across disciplines.

For: Modular high-fidelity MDO

Established | Python | NOASSERTION | Last push: 2026-10-16

OpenMDAO/mphys / Documentation

Selection note: It is an integration layer; individual examples require compatible external solvers and their dependencies.

OpenMDAO

Python framework for tightly coupled systems analysis and multidisciplinary optimization. Useful for connecting CFD, structures and engineering models while managing derivatives, nonlinear coupling and parallel model execution.

For: Coupled engineering optimization

Established | Python | NOASSERTION | Last push: 2026-10-22

OpenMDAO/OpenMDAO / Documentation

Selection note: CFD analysis components and reliable derivatives must be supplied; it does not replace a flow solver.

OpenTURNS

C++ and Python library for probabilistic modelling and uncertainty treatment in industrial studies. Useful for wrapping CFD evaluations, defining uncertain inputs and constructing reliability, sensitivity or surrogate analyses.

For: Probabilistic CFD study design

Established | C++ | NOASSERTION | Last push: 2026-10-24

openturns/openturns / Documentation

Selection note: Physics and solver interfaces remain application-specific; verify sampling accuracy and assumptions about input distributions.

pyOptSparse

Framework for defining and solving constrained nonlinear optimization problems with sparse derivatives. Useful as an optimization driver around aerodynamic CFD models and other expensive engineering analyses with many constraints.

For: Sparse constrained aerodynamic optimization

Established | Python | LGPL-3.0 | Last push: 2026-10-12

mdolab/pyoptsparse / Documentation

Selection note: Some optimizer backends require separate proprietary licenses; check the chosen backend and installation.

RAVEN

Simulation workflow framework for uncertainty quantification, optimization, model reduction and probabilistic risk analysis. Useful for orchestrating external CFD studies, sampling parameter spaces and training surrogates around expensive evaluations.

For: Simulation orchestration and risk/UQ studies

Established | Python | Apache-2.0 | Last push: 2026-10-24

idaholab/raven / Documentation

Selection note: Requires model interfaces, configuration and domain-specific validation; probabilistic output is only as reliable as its assumptions.

SALib

Python implementations of global sensitivity-analysis methods such as Sobol and Morris. Useful for ranking uncertain CFD inputs after sampling and evaluating a separate solver or surrogate model.

For: Global sensitivity of CFD outputs

Established | Python | MIT | Last push: 2026-10-18

SALib/SALib / Documentation

Selection note: Generates designs and analyzes outputs; the user supplies solver runs and checks estimator convergence.

UQpy

General Python toolbox for modelling uncertainty in physical and mathematical systems. Useful for uncertainty propagation and surrogate-based analyses around CFD models when sampling and probabilistic workflows need programmable control.

For: Programmable CFD uncertainty workflows

Established | Python | MIT | Last push: 2026-10-06

SURGroup/UQpy / Documentation

Selection note: A general UQ library rather than a CFD package; connect and validate the simulation model separately.

Numerical infrastructure & parallel computing

PDE assembly, adaptive grids, linear algebra and performance backbones used to build CFD applications.

11 repositories in this topic.

AMReX

Block-structured adaptive-mesh framework supplying parallel data structures, particles, embedded boundaries and CPU/GPU infrastructure for building CFD applications; it does not provide a single configured flow model.

For: Building adaptive parallel CFD applications

Established | C++ | NOASSERTION | Last push: 2026-10-23

AMReX-Codes/amrex / Documentation

Selection note: Application developers must supply governing equations, discretizations and physical models. Match the framework revision and accelerator configuration to the consuming solver.

deal.II

C++ adaptive finite-element library providing meshes, finite-element spaces, assembly and parallel algorithms for custom CFD programs, including documented flow examples rather than one universal ready-to-run solver.

For: Adaptive finite-element solver development

Established | C++ | NOASSERTION | Last push: 2026-10-25

dealii/dealii / Documentation

Selection note: A numerical development library; the application determines governing equations and stable discretization. Optional PETSc, Trilinos and p4est dependencies must be version-compatible.

DOLFINx / FEniCSx

C++ and Python finite-element environment for expressing and assembling PDE problems, useful for programmable Navier-Stokes studies; users still formulate weak forms, constraints, boundary conditions and solver strategies.

For: Programmable finite-element flow formulations

Established | C++ | LGPL-3.0 | Last push: 2026-10-24

FEniCS/dolfinx / Documentation

Selection note: DOLFINx is the FEniCSx computational component, not a standalone packaged CFD model. Follow documentation matching the installed release and compatible FEniCSx dependencies.

Ginkgo

C++ sparse linear-algebra library targeting many-core CPUs and GPUs, useful inside CFD pressure or coupled-system solvers; users provide matrices and algorithms appropriate to their discretized equations.

For: Portable GPU sparse-system backends

Established | C++ | BSD-3-Clause | Last push: 2026-10-25

ginkgo-project/ginkgo / Documentation

Selection note: A linear-algebra backend, not a CFD application. Benchmark complete solver convergence and data movement; accelerator support does not guarantee faster end-to-end simulation.

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hypre

Parallel sparse-system library featuring algebraic and structured multigrid methods, useful for CFD pressure and diffusion solves; it supplies numerical components rather than a turnkey fluid simulation.

For: Parallel multigrid pressure-system solution

Established | C | NOASSERTION | Last push: 2026-10-12

hypre-space/hypre / Documentation

Selection note: Select an interface and preconditioner appropriate to matrix structure and symmetry; GPU support varies by algorithm and build. Current licensing is Apache-2.0 OR MIT.

Kokkos Core

C++ performance-portability programming library abstracting parallel execution and memory layouts, useful for implementing CFD kernels across CPU and GPU architectures; it supplies no governing equations or flow solver.

For: Portable CFD kernel implementation

Established | C++ | NOASSERTION | Last push: 2026-10-23

kokkos/kokkos / Documentation

Selection note: Performance portability still requires algorithm, layout and backend choices. Check compiler and device support for the chosen release; numerical correctness remains the application's responsibility.

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libCEED

Library for factored finite-element operator evaluation across CPU and GPU backends, enabling matrix-free high-order CFD implementations; its algebraic interface integrates into applications rather than defining their complete physics.

For: Portable high-order operator kernels

Established | C | BSD-2-Clause | Last push: 2026-10-23

CEED/libCEED / Documentation

Selection note: Applications retain mesh, parallel decomposition and solver responsibilities. Backend behavior and deterministic capabilities vary; bundled fluid miniapps illustrate integration rather than certify arbitrary CFD models.

MFEM

Modular parallel finite-element library with high-order spaces, adaptive meshes and operator infrastructure for developing CFD discretizations; examples and miniapps are building blocks rather than complete engineering workflows.

For: High-order finite-element flow research

Established | C++ | BSD-3-Clause | Last push: 2026-10-25

mfem/mfem / Documentation

Selection note: Users select stable spaces, weak forms, boundary conditions and solvers. Reproducing an example does not establish the suitability of a newly assembled flow formulation.

p4est

Parallel forest-of-octrees library managing adaptive mesh refinement, coarsening and partitioning, useful beneath scalable CFD discretizations; the consuming application supplies numerical fluxes, physical fields and time evolution.

For: Distributed tree-based adaptive CFD meshes

Established | C | GPL-2.0 | Last push: 2026-10-19

cburstedde/p4est / Documentation

Selection note: Mesh infrastructure rather than a flow solver. Official guidance warns that auto-generated GitHub source archives lack required components; use maintained release tarballs or initialize dependencies correctly.

PETSc

Parallel numerical toolkit providing sparse algebra, nonlinear solvers, time integration and distributed meshes for CFD implementations; application code still supplies the flow equations, discretization and boundary conditions.

For: Scalable coupled CFD solver infrastructure

Established | C | NOASSERTION | Last push: 2026-10-24

petsc/petsc / Documentation

Selection note: Official GitHub mirror; primary development and issue tracking are on GitLab. Solver/preconditioner choices and nullspace handling must match the discretized problem.

SUNDIALS

Time-integration and nonlinear-solver libraries for differential and algebraic systems, useful for stiff chemistry and semidiscrete CFD equations; applications supply spatial operators, physical models and appropriate solver interfaces.

For: Stiff reacting-flow integration components

Established | C | BSD-3-Clause | Last push: 2026-10-24

llnl/sundials / Documentation

Selection note: Not a spatial CFD solver. Choose tolerances, Jacobians, vector backends and preconditioners for the actual stiffness and conservation requirements; keep package versions consistent.

Essential resources beyond GitHub

Canonical software homes, experimental datasets and reference portals are included separately when GitHub is not their primary distribution point. Check current access and use terms.

Gmsh

Geometry construction, mesh generation and post-processing through a desktop interface or scripting APIs, useful for building reproducible CFD meshes and preparing parametric geometry studies.

Canonical development is on ONELAB GitLab. The official site documents GPLv2-or-later terms with a linking exception and separate commercial licensing options.

https://gmsh.info/

Basilisk

Provides adaptive Cartesian PDE solvers, interface methods and a large collection of reproducible examples. Its native site combines source, documentation, tests and researcher-contributed simulation cases.

Canonical source is distributed through Darcs. Use the official project pages and source history rather than an arbitrary GitHub mirror.

https://basilisk.fr/

OpenFOAM (OpenCFD distribution)

Provides the OpenCFD distribution of OpenFOAM with solver, model, meshing and post-processing documentation. Identify this distribution explicitly when choosing compatible tutorials, extensions and case files.

GPLv3 distribution distinct from the OpenFOAM Foundation edition. Follow the official download and version-specific documentation; its development repositories moved to GitLab.com.

https://www.openfoam.com/

Palabos

Provides a lattice-Boltzmann framework for research and engineering flow applications. Its official site links the current source repository, documentation and examples covering interfaces, particles and porous media.

Official source is hosted on GitLab and distributed under AGPLv3. Applications generally require programming rather than a turnkey graphical interface.

https://palabos.unige.ch/

waLBerla

Combines lattice-Boltzmann flow modeling, particle dynamics, moving geometries and code generation in a parallel multiphysics framework. Useful for constructing coupled flow applications on structured block grids.

Official project states GPLv3; its maintained source is on the project GitLab server. Select models and generated kernels appropriate to the target hardware.

https://www.walberla.net/

OpenLB

Provides a modular lattice-Boltzmann library with examples spanning engineering flow, thermal transport, particles and porous media. The official user guide documents model setup and execution.

The official project overview states GPLv2 for code. Documentation has separate GNU Free Documentation License terms; use the official release package.

https://www.openlb.net/

TrioCFD

Documents the CEA turbulent-flow application built on TRUST, including thermal hydraulics, front tracking and fluid-structure models. Useful for understanding its model hierarchy and application scope.

Source-only distribution now hosted on GitLab. Follow the versioned official documentation and compile against the required TRUST environment.

https://gitlab.com/cea-trust-platform/triocfd/triocfd-code

OpenGeoSys

Models coupled thermal, hydraulic, mechanical and chemical processes in porous and fractured media. Offers subsurface-flow workflows, benchmark documentation and examples relevant to groundwater and geothermal research.

BSD-3-Clause project with active source hosted on its GitLab server. The old ufz/ogs GitHub repository explicitly directs users away from that stale location.

https://www.opengeosys.org/

Turbulence Modeling Resource

Documents RANS turbulence-model variants alongside verification grids, grid-convergence examples and fundamental validation cases. Helps distinguish implementation errors from the modeling limits of a turbulence closure.

Moved from NASA Langley to the TMBWG site in February 2026, as confirmed by NASA. Some large files have separate NASA-hosted download locations.

https://tmbwg.github.io/turbmodels/

ERCOFTAC Knowledge Base Wiki

Organizes CFD application challenges, underlying flow regimes and practice guidance. Useful for understanding a case context, reviewing modeling limitations and locating documented industrial or canonical comparisons.

ERCOFTAC advertises the wiki as free to visitors; individual test-data packages can have separate access and citation arrangements.

https://www.ercoftac.org/products_and_services/wiki/

ERCOFTAC Classic Collection

Indexes the original ERCOFTAC collection of flow test cases, experimental data and computational comparisons. Includes case descriptions and literature connections useful for selecting established validation exercises.

The official ERCOFTAC portal describes open access; inspect each dataset, its current hosting location and citation requirements.

https://www.ercoftac.org/products_and_services/classic_collection_database/

Johns Hopkins Turbulence Database

Offers queryable numerical turbulence fields through web services and data subsets. Useful for investigating velocity statistics, derivatives, particle trajectories and learning methods without storing complete simulations locally.

Follow dataset-specific citation and access instructions. Web-service limits, authentication requirements and storage formats vary; API clients are separate from the underlying data.

https://turbulence.idies.jhu.edu/home

Engine Combustion Network

Collects documented spray and engine experiments with common operating conditions, injector geometry and measured quantities. Supports comparative assessments of injection, mixing, ignition, flame stabilization and soot models.

Read each experiment's definitions, uncertainty, operating conditions and citation instructions. Matching a named Spray A or Spray G case requires more than reproducing nominal pressure and temperature.

https://ecn.sandia.gov/

University of Texas turbulence data

Provides downloadable DNS statistics and supporting information for turbulent channel and Couette flows. Useful for checking wall-scaled profiles, stresses and other canonical turbulence statistics.

Read the README accompanying each Reynolds-number dataset and cite the associated papers. These statistics describe specific DNS configurations rather than arbitrary wall-bounded flows.

https://turbulence.oden.utexas.edu/

Turbulent Non-Premixed Flames Workshop data

Collects experimental turbulent-flame datasets with scalar and velocity measurements in documented configurations. Supports comparisons of combustion submodels across piloted jets, bluff bodies, swirl and hydrogen flames.

Providers explicitly request reading documentation and referenced papers, including uncertainty information. The workshop does not guarantee the accuracy of every contributed dataset.

https://tnfworkshop.org/data-archives/

NASA NPARC verification and validation archive

Provides example, verification and validation cases associated with Wind-US, alongside broader CFD assessment guidance. Useful for learning how numerical checks and experimental comparisons answer different questions.

A maintained-access legacy archive with cases from different periods; check case dates, solver assumptions and source-data terms before reuse.

https://www.grc.nasa.gov/www/wind/valid/

Drag and Aeroelastic Prediction Workshops

Provides common aerodynamic geometries, experimental data, grids and comparison protocols for transonic drag and aeroelastic studies. Useful for separating mesh, numerical, turbulence-model and deformation effects.

Workshop participation and data use are different activities. Follow the selected edition's geometry, data, submission and citation requirements; do not mix results across configurations.

https://www.aiaa-dpw.org/

AIAA CFD High-Lift Prediction Workshop

Collects standardized high-lift aircraft test cases, geometry, grids and comparison results. Helps researchers assess difficult separated-flow predictions around deployed slats and flaps under shared conditions.

Use the correct workshop edition and configuration. Individual geometry and experimental files may carry their own attribution, distribution and interpretation requirements.

https://www.nasa.gov/reference/aiaa-cfd-high-lift-prediction-workshop/

Alphabetical project index

Click a project name to jump to its entry. Category labels describe the main placement; tags in the searchable directory cover overlaps.

ABLATE - Combustion, chemistry & reacting flows

ADCIRC - Ocean, atmosphere, wind & environmental flow

ADflow - General-purpose solvers & frameworks

ADIOS2 - Visualization, post-processing & I/O

AeroSandbox - Automation, optimization & uncertainty

AFiD - Turbulence, DNS, LES & HPC

AMReX - Numerical infrastructure & parallel computing

ANUGA - Ocean, atmosphere, wind & environmental flow

CadQuery - Geometry, CAD & mesh generation

CaNS - Turbulence, DNS, LES & HPC

Cantera - Combustion, chemistry & reacting flows

CFD Python - Verification, benchmarks & learning

CFDBench - Verification, benchmarks & learning

CFDEM coupling PUBLIC - Multiphase flow, droplets & sprays

CGAL - Geometry, CAD & mesh generation

CGNS - Visualization, post-processing & I/O

Chaospy - Automation, optimization & uncertainty

CM1 - Ocean, atmosphere, wind & environmental flow

code_saturne - General-purpose solvers & frameworks

COOLFluiD - General-purpose solvers & frameworks

DAFoam - Coupled physics & specialist applications

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DAPPER - Scientific ML, differentiable CFD & ROM

deal.II - Numerical infrastructure & parallel computing

Dedalus - Turbulence, DNS, LES & HPC

DeepXDE - Scientific ML, differentiable CFD & ROM

Delft3D - Ocean, atmosphere, wind & environmental flow

DOLFINx / FEniCSx - Numerical infrastructure & parallel computing

DualSPHysics - Lattice Boltzmann & particle methods

DuMux - Coupled physics & specialist applications

Elmer FEM - General-purpose solvers & frameworks

ERF - Ocean, atmosphere, wind & environmental flow

ExaDG - High-order & spectral methods

EZyRB - Scientific ML, differentiable CFD & ROM

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FiPy - General-purpose solvers & frameworks

Fire Dynamics Simulator - Combustion, chemistry & reacting flows

FLEXI - High-order & spectral methods

Fluidity - General-purpose solvers & frameworks

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GeoClaw - Ocean, atmosphere, wind & environmental flow

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Giverny / JHTDB access - Verification, benchmarks & learning

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hypre - Numerical infrastructure & parallel computing

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IBAMR - Coupled physics & specialist applications

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incflo - Turbulence, DNS, LES & HPC

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Kratos Multiphysics - General-purpose solvers & frameworks

Kynema-SGF (formerly AMR-Wind) - Ocean, atmosphere, wind & environmental flow

Kynema-UGF (formerly Nalu-Wind) - Ocean, atmosphere, wind & environmental flow

Lethe - General-purpose solvers & frameworks

Lettuce - Lattice Boltzmann & particle methods

libCEED - Numerical infrastructure & parallel computing

Mantaflow - Multiphase flow, droplets & sprays

MASA - Verification, benchmarks & learning

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MFC - Multiphase flow, droplets & sprays

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MIRGE-Com - High-order & spectral methods

MITgcm - Ocean, atmosphere, wind & environmental flow

MOM6 - Ocean, atmosphere, wind & environmental flow

MOOSE Navier-Stokes - General-purpose solvers & frameworks

MPAS-Model - Ocean, atmosphere, wind & environmental flow

MPhys - Automation, optimization & uncertainty

MRST - Coupled physics & specialist applications

NATriuM - Lattice Boltzmann & particle methods

Nek5000 - High-order & spectral methods

NekExamples - Verification, benchmarks & learning

nekRS - High-order & spectral methods

netCDF4-Python - Visualization, post-processing & I/O

Netgen - Geometry, CAD & mesh generation

NeuralOperator - Scientific ML, differentiable CFD & ROM

NVIDIA PhysicsNeMo - Scientific ML, differentiable CFD & ROM

Oceananigans.jl - Turbulence, DNS, LES & HPC

oomph-lib - Coupled physics & specialist applications

OpenFOAM Foundation - General-purpose solvers & frameworks

OpenMDAO - Automation, optimization & uncertainty

OpenPNM - Coupled physics & specialist applications

OpenSBLI - Turbulence, DNS, LES & HPC

OpenTURNS - Automation, optimization & uncertainty

OPM Flow - Coupled physics & specialist applications

p4est - Numerical infrastructure & parallel computing

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PDEBench - Verification, benchmarks & learning

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pygalmesh - Geometry, CAD & mesh generation

pyGeo - Geometry, CAD & mesh generation

pygmsh - Geometry, CAD & mesh generation

pyHyp - Geometry, CAD & mesh generation

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pyMARS - Combustion, chemistry & reacting flows

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pyOptSparse - Automation, optimization & uncertainty

PySINDy - Scientific ML, differentiable CFD & ROM

PySPH - Lattice Boltzmann & particle methods

PyVista - Visualization, post-processing & I/O

RAVEN - Automation, optimization & uncertainty

Reaction Mechanism Generator - Combustion, chemistry & reacting flows

Riemann Problems and Jupyter Solutions - Verification, benchmarks & learning

Riemann-Solvers - Verification, benchmarks & learning

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Transolver - Scientific ML, differentiable CFD & ROM

trimesh - Geometry, CAD & mesh generation

Trixi.jl - High-order & spectral methods

UCNS3D - High-order & spectral methods

UQpy - Automation, optimization & uncertainty

vedo - Visualization, post-processing & I/O

VisIt - Visualization, post-processing & I/O

VTK - Visualization, post-processing & I/O

WaterLily.jl - Turbulence, DNS, LES & HPC

WRF - Ocean, atmosphere, wind & environmental flow

Xcompact3d / Incompact3d - Turbulence, DNS, LES & HPC

yt - Visualization, post-processing & I/O